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Intellectual Property Protection Lost and Competition: An Examination Using Large Language Models

Contents

1 Big picture

- **Question.** What happens to firms, innovation, competition, and valuation when patent protection is unexpectedly weakened across broad technology areas?
- **Shock.** The paper studies the 2014 Supreme Court decision *Alice Corp. v. CLS Bank International*. Alice weakened patentability for business-method, software, data-processing, and related patents that implement abstract ideas without a sufficiently novel transformation.
- **Measurement challenge.** Whether any particular pre-Alice granted patent would lose protection after Alice is legally and textually complex. The paper solves this by using a large language model, ModernBERT, to score patent-level exposure to Alice.
- **Treatment.** A firm's exposure is the value-weighted share of its pre-2014 patent portfolio whose patents are predicted to become vulnerable after Alice. Treatment is continuous and normalized as a z-score.
- **Main result.** Alice reduces patenting overall, but its real effects are asymmetric: large exposed firms often gain, while small exposed firms lose.
- **Large firms.** Large exposed firms have higher sales growth and profitability, and they face fewer patent lawsuits. Their non-patent barriers to entry and organizational resources allow them to protect rents even when patent protection weakens.
- **Small firms.** Small exposed firms face more VC-backed entry, higher product-market encroachment, more competition complaints, lower profit margins, lower market-to-book valuations, and greater use of R&D and nondisclosure agreements.
- **Mechanism.** Alice weakens a legal barrier that especially protected smaller firms' product niches. Larger firms have other defenses, such as scale, market share, asset specificity, and capital.
- **Economic takeaway.** Strong patent protection is not uniformly pro- or anti-competitive. It can protect smaller firms from large-firm encroachment, while its removal can reallocate rents toward large incumbents.

2 Incremental contribution of the paper

- **From single-patent invalidation to technology-area IP loss.** Prior studies often examine the loss of protection for individual patents. This paper studies a broad legal shock that weakens current and future patent protection in entire technology spaces.
- **Firm-level exposure using LLMs.** The paper contributes a ModernBERT-based method for converting long patent text into firm-level exposure to a legal doctrine. This is an important measurement innovation because manual classification is infeasible for hundreds of thousands of patents.
- **Heterogeneous treatment effects by firm size.** The central empirical contribution is not merely that weaker IP protection changes patenting; it shows that firm size sharply mediates the sign of the effect on competition and value.
- **Linking IP law to product-market structure.** The paper connects patent invalidation to VC entry, TNIC product-market encroachment, competition language in 10-Ks, lawsuits, NDAs, sales, margins, and valuations.
- **A new view of patents as small-firm protection.** The results suggest that patents can function as barriers protecting small innovators from large incumbents, rather than only as monopoly rights that stifle entry.
- **Validation of legal-text machine learning.** The paper validates Alice scores using USPTO post-grant reviews and shows that ModernBERT scores strongly predict whether Alice is cited in patent challenges.

- **Policy contribution.** The paper implies that weakening patent protection can increase competition in some dimensions, but may also increase market concentration if large firms are better able to exploit lost protection.

3 Data and measurement

3.1 The Alice shock and affected technology areas

- Alice weakened patents that claim abstract ideas implemented using generic computer or business methods.
- Affected technologies include data processing, commerce, finance, payment systems, games, information retrieval, computer security, network applications, and testing in microbiology and enzymology.
- Table 1 shows that many affected categories had large declines in patent applications after 2013. Finance falls by 51.9%, commerce data-processing methods by 34.7%, and coin-freed facilities by 32.3% from 2013 to the 2015–2017 average.
- Some categories, such as network security, do not decline, showing that the shock is heterogeneous across technology areas.

Interpretation: Alice is useful empirically because it is a nationwide legal shock, not a firm-specific policy. The challenge is measuring which firms' pre-existing patents were actually exposed.

3.2 Patent universe and Alice-rejected training data

- The raw universe contains more than 3.8 million patents granted between 1994 and the Alice decision date, June 19, 2014.
- To make prediction tractable, the authors focus on patents with primary CPC codes also used by USPTO applications rejected under Alice.
- This filtering yields 642,678 pre-Alice granted patents to be scored.
- The positive training class consists of 33,734 unique patent applications explicitly rejected under Alice.
- Negative examples are granted patents in the same CPC areas, sampled to maintain comparable technology composition.

Interpretation: The prediction task is not simply about text similarity. It asks whether patent claims are semantically similar to patents rejected under a legal doctrine.

3.3 ModernBERT classification

- The paper uses ModernBERT, a long-context transformer encoder that can process up to 8192 tokens.
- Older BERT-style models are limited to 512 tokens, which is often too short for patent descriptions.
- ModernBERT is fine-tuned on Alice-rejected and granted patents and used to estimate each patent's probability of losing protection.
- The final ModernBERT score is the average of two high-performing training designs: one optimized for F1 score and the other for accuracy.

Interpretation: The LLM allows the paper to score legal exposure at scale while preserving contextual meaning in long patent texts.

3.4 Firm-level treatment variable

For each firm, the baseline treatment variable is value weighted:

$$Treatment_i = \frac{\sum_{p \in P_i} PatentValue_{ip} \times AliceScore_p}{Sales_i}, \tag{1}$$

where P_i is the set of firm i's pre-Alice patents. The paper also uses a citation-weighted treatment measure. All treatment variables are normalized as z-scores:

$$ZTreatment_i = \frac{Treatment_i - \overline{Treatment}}{sd(Treatment)}. \tag{2}$$

Interpretation: Treatment is continuous because firms differ in how much of their patent portfolio is exposed to Alice. A one-unit coefficient can be interpreted as the effect of a one-standard-deviation increase in exposure.

3.5 Firm sample and outcomes

- The main public-firm panel contains 19,808 firm-year observations and 3,646 firms from 2011 to 2017, excluding 2014 as the treatment year.
- Firms are split into small and large groups based on whether their assets are below or above the median assets of their TNIC peers in 2013.
- Innovation outcomes include patent applications scaled by assets and sales, R&D scaled by sales, and patent lawsuit outcomes.
- Competition outcomes include VC-funded entry into the firm's product market, competition language in 10-Ks, and TNIC product-market encroachment.
- Performance outcomes include sales growth, gross profit margin, and market-to-book ratio.
- Legal substitution outcomes include NPE and operating-company lawsuits and nondisclosure-agreement language in 10-K filings.

Interpretation: The outcome set is broad because the paper asks how weakened patent protection changes both innovation incentives and market structure.

4 Empirical specifications

4.1 ModernBERT performance metrics

The paper evaluates models using precision, recall, F1 score, and accuracy:

$$Precision = \frac{TP}{TP + FP}, \qquad Recall = \frac{TP}{TP + FN}, \tag{3}$$

$$F1 = 2\frac{Precision \times Recall}{Precision + Recall}, \qquad Accuracy = \frac{TP + TN}{TP + TN + FP + FN}. \tag{4}$$

Interpretation: F1 emphasizes performance on the rejected-patent class, while accuracy measures overall classification success. The paper averages two ModernBERT models because one maximizes F1 and the other maximizes accuracy.

4.2 Post-grant review validation

The validation regression is a logit model:

$$Pr(AliceReferenced_p = 1) = \Lambda\left(\alpha + \beta AliceScore_p + X_p'\gamma\right), \tag{5}$$

where $AliceReferenced_p$ equals one when a post-grant review cites Alice.

Interpretation: A positive coefficient on ModernBERT score indicates that the LLM score predicts later legal challenges invoking Alice, providing out-of-sample legal validation.

4.3 Main continuous-treatment difference-in-differences

For firm outcome Y_{it}, the main specification is:

$$Y_{it} = \alpha_i + \gamma_t + \beta_S(Small_i \times Post_t \times Treatment_i) + \beta_L(Large_i \times Post_t \times Treatment_i) + \epsilon_{it}. \tag{6}$$

- α_i are firm fixed effects.
- γ_t are year fixed effects.
- $Post_t$ equals one after the Alice decision.
- $Small_i$ and $Large_i$ identify size groups.
- The main tests compare β_S and β_L.

Interpretation: The specification asks whether outcomes change after Alice more strongly for firms whose patent portfolios were more exposed, separately for small and large firms.

4.4 Entropy balancing

Entropy balancing constructs weights so that treated and control observations match on pre-treatment characteristics:

- two-digit SIC industry;
- 2013 sales;
- 2013 assets;
- log age;
- total number of valid patents as of 2013.

Interpretation: Because treatment is continuous and related to firm characteristics, balancing is important. Table 6 shows that strong pre-balancing differences disappear after entropy weights are applied.

4.5 Pairwise product-market encroachment

For firm pair (i, j), the paper estimates changes in product-market similarity:

$$\Delta TNICScore_{ij,t} = \alpha_{ij} + \gamma_t + \beta(Treat_{ij} \times Post_t) + \epsilon_{ij,t}. \tag{7}$$

A more detailed version separates pairs by whether the treated firm is small or large.

Interpretation: This specification measures whether rivals become more similar after Alice, which would indicate product-market encroachment into previously protected niches.

4.6 Structural-break and pre-trend tests

The paper estimates yearly interactions:

$$Y_{it} = \alpha_i + \gamma_t + \sum_{k \neq 2013} \beta_k (Small_i \times Treatment_i \times 1\{t = k\}) + \epsilon_{it}. \tag{8}$$

The 2011–2013 coefficients define a pre-trend projection, and post-2014 effects are compared to that projection.

Interpretation: The structural-break figures show whether the post-Alice divergence is genuinely new rather than the continuation of a pre-existing trend.

5 Identification and empirical design

5.1 Source of variation

- The shock is the Supreme Court's Alice decision, which was broad, national, and not chosen by individual firms.
- Cross-sectional treatment intensity comes from pre-existing patent portfolios.
- The main identifying comparison is between more- and less-exposed firms before and after Alice, with firm and year fixed effects.
- The key heterogeneity is between small and large firms.

Interpretation: The paper uses the legal shock as the time-series variation and LLM-predicted patent exposure as the cross-sectional intensity.

5.2 Why the shock is plausibly unexpected

- Before Alice, lower courts disagreed about the patentability of the relevant business-method and software-related claims.
- The decision was uncertain and was not a firm-specific event.
- It affected broad technology classes and future patentability, not only one firm's challenged patent.

Interpretation: The shock is more plausibly exogenous than a firm's own litigation loss because it comes from a Supreme Court interpretation of patent eligibility.

5.3 Main threats and responses

- **Treatment correlated with firm size and industry:** entropy balancing and firm fixed effects address observable imbalance.
- **Pre-existing trends:** structural-break figures and pre-trend tests show little evidence of differential pre-trends.
- **Other patent cases:** the paper compares Alice with Bilski, Mayo, and Myriad and finds Alice is the central shock for the studied technology areas.
- **Fuzzy treatment measurement:** ModernBERT is validated using holdout performance and post-grant reviews.
- **Parallel-trends violations:** robustness uses nearest-neighbor matching and Rambachan-Roth style sensitivity analysis.

Interpretation: Identification is strongest when the same pattern appears across patenting, competition, valuation, lawsuits, and textual evidence.

6 Tables and figures

6.1 Table 1 and Figure 1: Alice shock and LLM measurement pipeline

Read:

- Table 1 shows sharp post-Alice declines in patent applications in several affected industries, especially finance, commerce, and coin-freed facilities.
- Figure 1 shows the construction of the training and testing sets for ModernBERT.
- The paper starts with 33,734 Alice-rejected applications and a large pool of granted patents in related CPC categories.
- Training sets vary the granularity of matched accepted patents, from same CPC main group to broader CPC categories.

Economic message: The table documents that Alice mattered for patenting, and the figure shows how the paper turns legal rejection data into scalable firm-level treatment measures.

Table 1
Annual patent applications and post-alice rejections by industry.
This table reports annual statistics from USPTO patent applications and the corresponding percentage that were rejected in parentheses based on the Supreme Court's Alice decision for the top 12 industries with patent rejections. The rejection data provided by Lu et al. (2017) extends until 2016; therefore, the ratio of rejection is assigned NA for 2017. *Change* reports the percentage change from the number of patent applications in 2013 to the average number of patent applications for the 2015–2017 period. Corresponding CPCs for each industry are provided in Table IA5.

Patent Applications and USPTO Alice Rejections — Top 12 industries									
Number of Patent Applications & Rejection Percentage									
Industry	2008–2009	2010–2011	2012	2013	2014	2015	2016	2017	Change (2013 to 2015–2017)
Commerce (Data Processing Methods)	6582 (11.7%)	7675 (17.9%)	5032 (29.8%)	5564 (36.2%)	5221 (23.2%)	4246 (6.6%)	3406 (1.5%)	3253 (NA)	−34.7%
Administration (Data Processing Methods)	6681 (6.7%)	6250 (11.1%)	3658 (20.8%)	2958 (31.3%)	2969 (16.7%)	2500 (3.6%)	2531 (0.6%)	2574 (NA)	−14.3%
Finance (Data Processing Methods)	2297 (9.4%)	2662 (13.2%)	1545 (22.5%)	1752 (42.1%)	1512 (37.8%)	1038 (8.7%)	775 (1.9%)	715 (NA)	−51.9%
Payment Systems (Data Processing Methods)	1603 (9.9%)	2043 (12.9%)	1673 (26.6%)	1946 (36.7%)	2182 (24.4%)	2158 (5.8%)	2029 (1.9%)	1902 (NA)	4.3%
Coin-freed Facilities or Services (Coin-freed or Like Apparatus)	2385 (3.9%)	1665 (6.8%)	1221 (17.0%)	1407 (34.3%)	1134 (31.2%)	980 (14.9%)	939 (6.7%)	938 (NA)	−32.3%
Information Retrieval (Digital Data Processing)	7981 (0.5%)	8451 (1.2%)	5850 (2.4%)	6566 (4.1%)	6650 (5.1%)	6339 (2.0%)	6196 (1.0%)	5818 (NA)	−6.8%
Video Games (Games)	1414 (4.5%)	1504 (7.0%)	919 (12.5%)	1045 (27.4%)	1010 (19.6%)	781 (7.8%)	847 (3.4%)	931 (NA)	−18.4%
Specialized For Sectors (Data Processing Methods)	515 (4.9%)	918 (10.9%)	753 (15.5%)	845 (32.1%)	881 (19.3%)	669 (4.5%)	848 (0.6%)	809 (NA)	−8.2%
Computer Security (Digital Data Processing)	3886 (1.6%)	3926 (1.5%)	2617 (2.6%)	2684 (5.0%)	2641 (5.2%)	2604 (4.0%)	2675 (0.7%)	2881 (NA)	1.3%
Network Security (Transmission of Digital Information)	3522 (0.8%)	3208 (0.8%)	2206 (1.9%)	2864 (4.3%)	3433 (5.5%)	4042 (3.4%)	4126 (0.8%)	3822 (NA)	39.5%
Network Specific Applications (Transmission of Digital Information)	3389 (0.8%)	3441 (1.5%)	2282 (3.2%)	2891 (6.0%)	3174 (4.7%)	3172 (2.2%)	3099 (0.6%)	2420 (NA)	0.2%
Measuring or Testing Processes (Microbiology & Enzymology)	3759 (1.3%)	4311 (2.5%)	2237 (4.3%)	2356 (4.9%)	2336 (3.2%)	2105 (2.6%)	2101 (0.6%)	2087 (NA)	−11.0%

Table 1: patent applications and Alice rejections

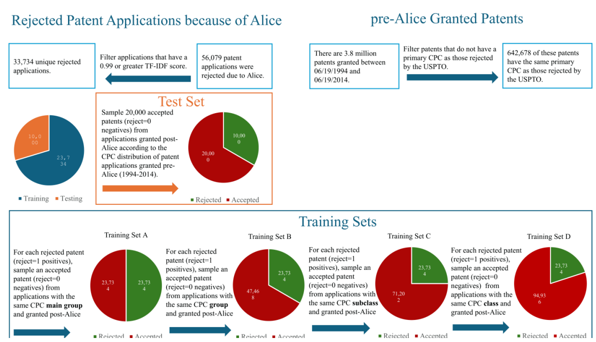

Figure 1: ModernBERT training and evaluation

6.2 Tables 2 and 3: ModernBERT performance and legal validation

Read:

- Table 2 shows that ModernBERT outperforms Longformer, SciBERT, BERT, RoBERTa, TF-IDF, and Word2Vec baselines.
- The ModernBERT ensemble has the highest F1 score, 0.700, and accuracy, 0.825.
- Table 3 shows that ModernBERT Alice scores strongly predict whether a post-grant review cites Alice.
- The score coefficient ranges from 5.422 to 7.576 and is significant at the 1% level across specifications.

Economic message: The measurement strategy is credible because the LLM predicts both held-out Alice rejections and later legal challenges that explicitly reference Alice.

Comparison of predictions for ModernBERT vs. other models.

This table compares predictions of ModernBERT (Warner et al., 2025), Longformer (Beltagy et al., 2020), SciBERT (Beltagy et al., 2019), BERT (Devlin et al., 2019), RoBERTa (Liu et al., 2019), TF-IDF (Robertson, 2004) and Word2Vec models (Mikolov et al., 2013) based on F_1 Score and Accuracy. F_1 Score is the harmonic mean of recall and precision, which are defined in Eq. (1). Accuracy is the ratio of correctly predicted observations to the total observations. For all models, we conduct four experiments in which the only difference is the way we create the training samples. In experiment A, for each of the 23,734 positives (rejected patent applications), we find a matching negative (a patent application that is granted) that is in the same CPC Main Group. In sample B, C, and D, without replacement, we keep adding 23,734 more matching *granted* patents to the negatives pool based on CPC Group, Subclass, and Class, respectively. Therefore, from A to D, each sample has 23,734 more negatives, i.e. granted patents, but the newly added ones are selected from a broader CPC. In the last column of the table, we use an ensemble of the two models that have the highest F_1 Score and Accuracy by taking the average of their prediction scores. The use of this ensemble is motivated by the fact that A typically has the highest F_1 Score and D has the highest Accuracy. For the testing, we only use applications and granted patents that were not used in the training. In the testing, we have 10,000 positives/rejected=1 in the sample of applications and for each positive, we choose two negatives/rejected=0. This 1:2 positives to negatives ratio is consistent with the expected rejection ratio having more negatives than positives, while not overestimating accuracy for models that do not learn but only predict negative outcomes. From the negatives pool, we thus sample 20,000 negatives 1000 times and boot-strap performance of the models. The table below then reports the average F_1 Score and Accuracy for each model.

Model name	A		B		C		D		$\frac{A+D}{2}$	
	F_1 Score	Accuracy	F_1 Score	Accuracy	F_1 Score	Accuracy	F_1 Score	Accuracy	F_1 Score	Accuracy
ModernBERT Finetune	0.677	0.766	0.653	0.797	0.655	0.811	0.662	0.815	0.700	0.825
Longformer Finetune	0.647	0.745	0.623	0.765	0.618	0.784	0.639	0.800	0.682	0.804
SciBERT Finetune	0.651	0.735	0.634	0.749	0.632	0.767	0.638	0.777	0.669	0.778
BERT Finetune	0.623	0.733	0.598	0.739	0.614	0.764	0.624	0.774	0.642	0.775
RoBERTa Finetune	0.600	0.716	0.555	0.740	0.540	0.756	0.515	0.758	0.592	0.765
TF-IDF + Logistic Regression	0.547	0.643	0.599	0.634	0.613	0.670	0.550	0.719	0.559	0.679
TF-IDF + Decision Tree	0.503	0.602	0.554	0.552	0.558	0.584	0.491	0.690	0.409	0.697
TF-IDF + Random Forest	0.628	0.743	0.368	0.717	0.263	0.696	0.209	0.689	0.387	0.723
Word2Vec + Logistic Regression	0.606	0.731	0.418	0.732	0.377	0.732	0.358	0.730	0.497	0.755
Word2Vec + Decision Tree	0.492	0.607	0.456	0.645	0.461	0.687	0.461	0.702	0.365	0.707
Word2Vec + Random Forest	0.619	0.747	0.439	0.746	0.387	0.739	0.365	0.735	0.500	0.766

Table 2: model comparison

Post-grant review prediction.

The table displays the results from a logit regression in which the dependent variable, Alice Referenced Dummy, is a dummy equal to one if the given patent in the post-grant review references the Alice case and zero otherwise. The sample consists of 391 unique patents from September 16, 2012 (the start date of post-grant reviews) to September 16, 2024, that experienced post-grant reviews. We note that 75 of these reviews specifically referenced the Alice case when requesting invalidation. ***Alice ModernBERT Score*** is the predicted value of Alice rejection from the ModernBERT model. ***Log(Number of Claims)*** is the logged number of claims of the patent. ***Petitioner Types*** indicate the type of entity petitioning for post-grant review. These include defensive aggregators, government, large operating companies, small or medium enterprises, unified patents, or other. The *t*-statistics are reported in parentheses; *, **, and *** denote the significance at the level of 10%, 5%, and 1%.

Dependent variable:	Alice referenced dummy			
	(1)	(2)	(3)	(4)
Alice ModernBERT Score	5.422***	5.399***	5.843***	7.576***
	(8.58)	(8.51)	(7.42)	(4.38)
Log(Number of Claims)		−0.257	−0.139	−1.448**
		(−0.89)	(−0.38)	(−2.31)
Observations	391	391	339	223
Petitioner Type	NO	NO	YES	YES
Filing Year Fixed Effect	NO	NO	YES	YES
Grant Year Fixed Effect	NO	NO	YES	YES
CPC Fixed Effect	NO	NO	NO	YES
Psuedo R^2	0.2508	0.2528	0.3641	0.5553

Table 3: post-grant review validation

6.3 Figure 2, Figure 3, and Table 4: treatment distribution and sample summary

Read:

- Figure 2 shows that high-Alice-score patent applications and grants become less common after Alice.
- Figure 3 shows that many firms have zero or near-zero exposure, but a right tail of firms has high Alice exposure.
- Table 4 describes the main public-firm sample: 19,808 observations and 3,646 firms.
- Post-Alice means show lower patent applications, lower lawsuit counts, higher VC entry, and more nondisclosure language.

Economic message: The shock is concentrated in a meaningful minority of patents and firms, creating cross-sectional treatment intensity.

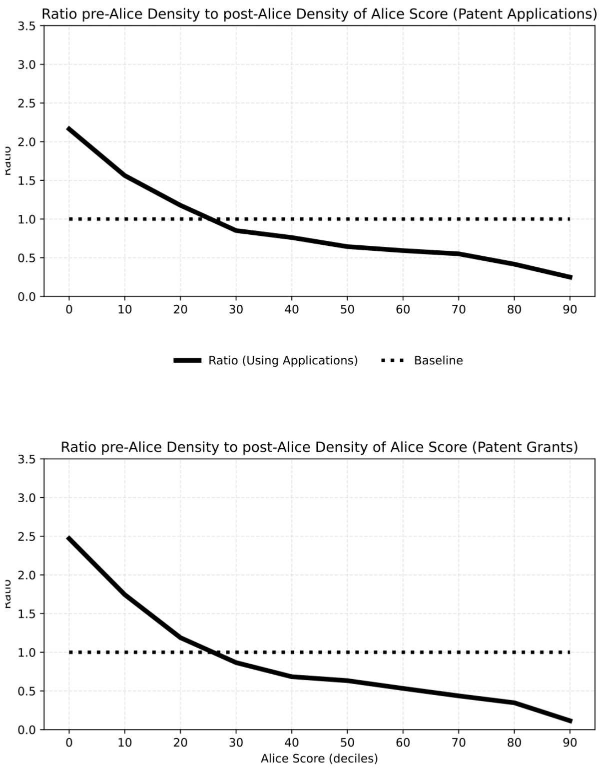

Figure 2: density shifts in Alice scores

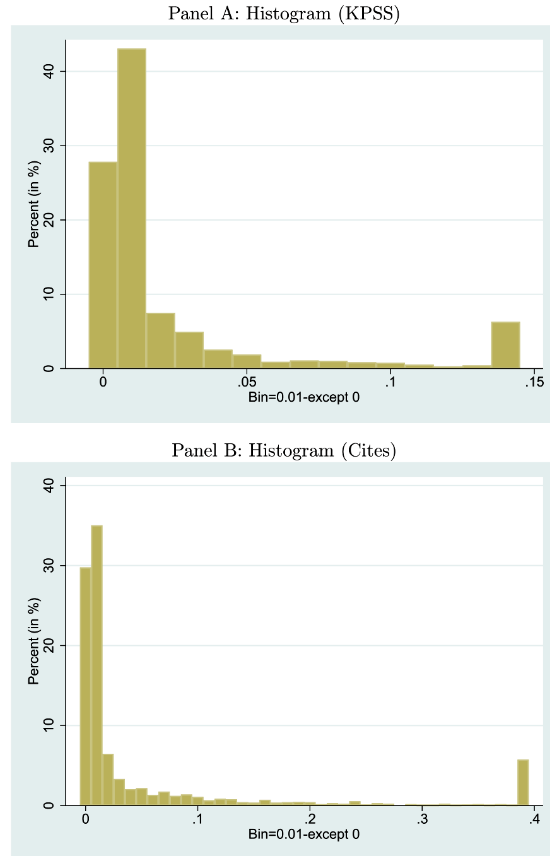

Figure 3: treatment distribution

Firm summary statistics.

This table provides summary statistics for our sample of public firms based on annual firm observations from 2011 to 2017 (excluding 2014, the treatment year). All variables are described in detail in the variable list in Appendix A and in Section 3 of the paper. *, **, and *** denote significant difference of the mean post-Alice vs. pre-Alice at the 10%, 5% and 1% level.

Variable	N	# Firms	Pre-Alice			Post-Alice			Diff
			Median	Mean	Std. Err.	Median	Mean	Std. Err.	
Panel A: Firm Characteristics									
Assets (in mil.)	19 808	3646	830.133	9139.905	1226.523	1251.855	11 393.070	1395.515	
Sales (in mil.)	19 808	3646	384.894	3509.421	253.765	551.892	4031.148	273.913	
Sales Growth	19 808	3646	0.064	0.088	0.003	0.037	0.033	0.003	***
Age	19 808	3646	18.000	21.347	0.265	22.000	25.510	0.280	***
Gross Profit Margin	19 808	3646	0.434	0.446	0.007	0.403	0.381	0.011	***
M/B Ratio	19 309	3603	1.133	1.906	0.040	1.187	1.756	0.031	***
Panel B: Innovation & Lawsuit Characteristics									
Treatment	19 808	3646	−0.306	0.001	0.017	−0.306	0.001	0.017	
R&D$_t$/Sales$_{t-1}$	19 808	3646	0.000	0.143	0.008	0.000	0.158	0.011	***
Patent App./Sales$_{t-1}$	19 808	3646	0.000	0.015	0.001	0.000	0.008	0.000	***
Patent App./Assets$_{t-1}$	19 808	3646	0.000	0.008	0.000	0.000	0.004	0.000	***
# NPE Alleged	19 808	3646	0.000	0.163	0.008	0.000	0.154	0.007	
# OC Alleged	19 808	3646	0.000	0.101	0.005	0.000	0.048	0.003	***
# Alleged	19 808	3646	0.000	0.280	0.012	0.000	0.227	0.010	***
Panel C: Competition Measures									
VCF/Sales$_{t-1}$	19 725	3646	0.014	0.122	0.005	0.012	0.195	0.012	***
Complaints	19 727	3646	13.484	14.586	0.113	13.611	14.709	0.116	
Nondisclose	18 486	3509	0.000	0.012	0.001	0.000	0.016	0.001	***

6.4 Tables 5 and 6: examples and entropy balancing

Read:

- Table 5 lists examples of high-Alice-score patents held by large and small companies.
- Large firms such as JPMorgan, Oracle, GE, and Disney have many other patents and larger patent portfolios.

- Small firms such as Aspira, Multimedia Games, Guidewire, and Stamps.com have fewer patents and smaller market values.
- Table 6 shows that entropy balancing removes strong pre-treatment differences in sales, assets, age, and patent counts.

Economic message: The examples illustrate why firm size matters: the same type of patent shock is absorbed very differently by firms with different product portfolios and barriers to entry.

Table 5
Sample of alice-impacted patents and owner characteristics.
This table provides a representative selection of patents exhibiting a high Alice ModernBERT score, accompanied by firm characteristics pertaining to the patent owners. Panel A presents data concerning large firms within our sample classification, while Panel B offers corresponding information for small firms. After the first column (company name), the next four columns contain information about the company's most-exposed patent to Alice (as indicated by the high Patent Alice ModernBERT Score). The final three columns display firm characteristics summarizing patenting intensity and the value of discounted total patent portfolio.

Company name	Patent number	Patent description	Patent Alice-score	KPSS disc. patent value (2014 $M)	Firm's total number of patents	Firm's # of alice-impacted patents	Firm's total patent value (2014 $M)
Panel A: Large Companies							
JPMORGAN CHASE & CO	6 502 080	A method of predicting a net reserve for a vehicle leased by a lessee from a lessor in accordance with a lease.	0.99	44.06	1,702	380	123,594
ORACLE CORP	8 738 489	A computer-implemented method that manages financial close schedules by creating, validating, duplicating, and setting parameters.	0.95	46.91	6,359	596	102,589
GENERAL ELECTRIC CO	8 751 423	A turbine performance diagnostic system that creates a performance report for one or more turbines.	0.93	22.17	6,359	908	44,037
WALT DISNEY CO	8 745 165	A system and method for managing distribution of rich media content.	0.93	69.19	1,640	114	23,960
Panel B: Small Companies							
ASPIRA WOMEN'S HEALTH INC (a.k.a, VERMILLION INC)	8 664 358	Methods to predict ovarian cancer presence, subtype, stage, treatment efficacy, and cancer development using diagnostic systems.	0.97	1.677	201	15	13.49
MULTIMEDIA GAMES HOLDING CO	8 708 806	A method that controls free plays in wagering games through a repeatable secondary game awarding additional plays.	0.90	9.44	308	68	206.67
GUIDEWIRE SOFTWARE INC	8 566 131	A system that manages insurance policy processing by enforcing holds, blocking operations, and releasing holds when criteria are met.	0.99	17.71	14	10	166.96
STAMPS.COM INC	8 612 361	System and method for handling payment errors with respect to delivery services	0.97	5.73	403	45	111.01

Table 5: examples of exposed patents

Table 6
Treatment and control group balance.
The table displays the results from regressions where key characteristics, both equal-weighted and entropy-weighted, are regressed on our Alice treatment variable. The goal is to illustrate the degree of imbalance prior to entropy weighting (expect significant results) and the efficacy of entropy balancing after weighting (expect insignificant results). In the odd columns, dependent variables are equal-weighted sales, assets, logged age, and the total number of valid patents until that year. In the even-numbered columns, the dependent variables are entropy-weighted using weights from the entropy balancing approach explained in Section 3.7. In columns (1)–(2) and (5)–(6), the data only includes observations from 2013 (i.e., the year before the Alice decision). All years are included in the other columns. All variables are described in Appendix A. Even-numbered regressions include year fixed effects, and standard errors are clustered at the firm level. t-statistics are reported in parentheses; *, **, and *** denote the significance at the level of 10%, 5%, and 1%.

PANEL A:	(1) Sales	(2) Weighted sales	(3) Sales	(4) Weighted sales	(5) Assets	(6) Weighted assets	(7) Assets	(8) Weighted assets
Treatment	−307.760***	0.030	−312.214***	0.041	−631.121***	0.052	−674.716***	0.115
	(−2.66)	(0.25)	(−4.59)	(0.33)	(−2.99)	(0.22)	(−4.66)	(0.40)
Sample	Only 2013	Only 2013	All Years	All Years	Only 2013	Only 2013	All Years	All Years
Year Fixed Effects	NO	YES	NO	YES	NO	YES	NO	YES
Observations	3646	3646	19 808	19 808	3646	3646	19 808	19 808
Adj. R^2	0.002	−0.000	0.003	−0.000	0.002	−0.000	0.007	0.000
PANEL B:	**(1) Log(Age)**	**(2) Weighted log(Age)**	**(3) Log(Age)**	**(4) Weighted log(Age)**	**(5) Patent count**	**(6) Weighted patent count**	**(7) Patent Count**	**(8) Weighted patent count**
Treatment	−0.047***	0.001	−0.050***	0.000	212.629***	0.003	230.970***	0.006
	(−3.58)	(0.01)	(−5.32)	(0.05)	(12.33)	(0.31)	(7.63)	(0.63)
Sample	Only 2013	Only 2013	All Years	All Years	Only 2013	Only 2013	All Years	All Years
Year Fixed Effects	NO	YES	NO	YES	NO	YES	NO	YES
Observations	3646	3646	19 808	19 808	3646	3646	19 808	19 808
Adj. R^2	0.003	0.000	0.030	0.001	0.040	−0.000	0.044	0.000

Table 6: treatment-control balance

6.5 Figure 4 and Table 7: innovation effects

Read:

- Table 7 shows that Alice reduces patent applications for both small and large firms.
- The effect is stronger for small firms using the KPSS value-based treatment measure.
- For patents scaled by sales, a one-standard-deviation increase in treatment reduces small-firm patenting by about 31% relative to the small-firm pre-Alice mean.
- Large-firm patenting falls by about 14% relative to the large-firm pre-Alice mean.
- Figure 4 shows no clear pre-trend and a sharp post-Alice divergence between small and large firms.

Economic message: Weakened IP protection reduces future patenting, especially for smaller firms whose patent portfolios are more central to product-market protection.

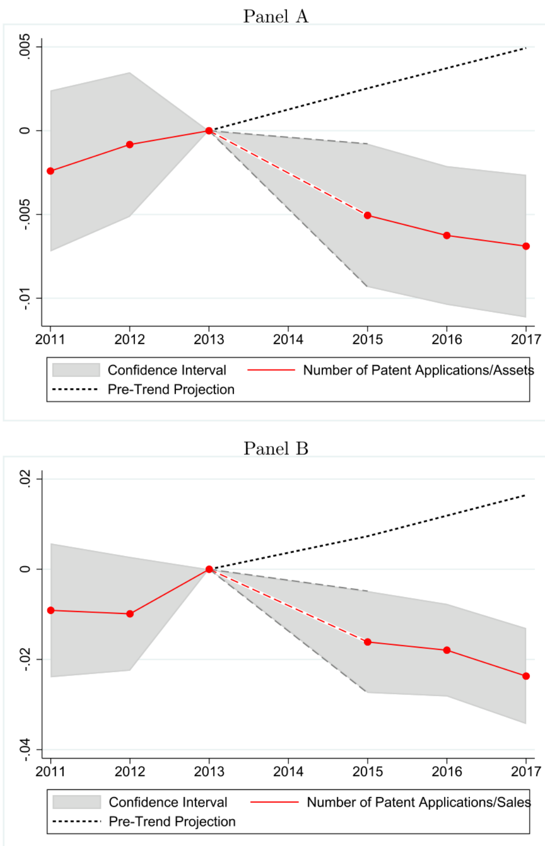

Figure 4: patenting divergence

Effects of alice on patent applications (ModernBERT).

The table displays the results from entropy-balanced panel data regressions in which patent applications are scaled by assets and sales as dependent variables. In columns (1)–(2), the dependent variable is the number of patent applications in year *t* divided by assets in year *t-1*; and in columns (3) and (4), the dependent variable is the number of patent applications in year *t* divided by sales in year *t-1*. *Treatment* is the relative impact of the Alice decision on the firm's patent portfolio measured using KPSS dollar values or citations scaled by sales (see Eq. (2) for the formula). The *Treatment* variable was normalized by subtracting the mean and dividing by the standard deviation, resulting in a z-score transformation. In the odd- and even-numbered columns, respectively, we use the KPSS and the number of citations approach to compute the *Patent Value* for the treatment variable. *Small* is a binary variable equal to one if a firm's total assets are less than the median total asset of its TNIC peers in 2013 and is zero otherwise. *Large* is 1-*Small*. *Post* is a dummy variable that equals one if the year is after the Alice decision and zero otherwise. *Small-Large Difference* is the difference between the estimates of Small × Post × Treatment and Large × Post × Treatment. All variables are described in detail in the variable list in Appendix A. All regressions include firm and year fixed effects. Standard errors are clustered at the firm level. *t*-statistics are reported in parentheses; *, **, and *** denote the significance at the level of 10%, 5%, and 1%.

Dependent variable:	$\frac{\#of\,PatentApp.}{Assets_{t-1}}$		$\frac{\#of\,PatentApp.}{Sales_{t-1}}$	
	(1)	(2)	(3)	(4)
Small × Post × Treatment	−0.006***	−0.004***	−0.015***	−0.008***
	(−7.32)	(−4.91)	(−9.79)	(−4.69)
Large × Post × Treatment	−0.002***	−0.003**	−0.002**	−0.004**
	(−2.71)	(−1.99)	(−2.07)	(−2.26)
Difference	−0.005***	−0.001	−0.013***	−0.004
	(−4.54)	(−0.84)	(−6.54)	(−1.49)
Observations	19 808	19 808	19 808	19 808
Firm Fixed Effects	YES	YES	YES	YES
Year Fixed Effects	YES	YES	YES	YES
Treatment Calculation	KPSS	Citation	KPSS	Citation
Adj. R^2	0.119	0.088	0.107	0.071

Table 7: patent applications

6.6 Figure 5, Table 8, and Table 9: competition and encroachment

Read:

- Table 8 shows that small exposed firms experience more VC-backed entry, more competition complaints, and higher R&D intensity after Alice.
- Large firms do not show comparable increases in competition measures.
- Figure 5 confirms a post-Alice structural break in VC entry and competition complaints for small versus large firms.
- Table 9 shows that product-market encroachment rises when small firms are in the treated pair.

Economic message: Alice makes small-firm product markets more contestable. Small firms respond by raising R&D and by facing greater entry and product overlap from rivals.

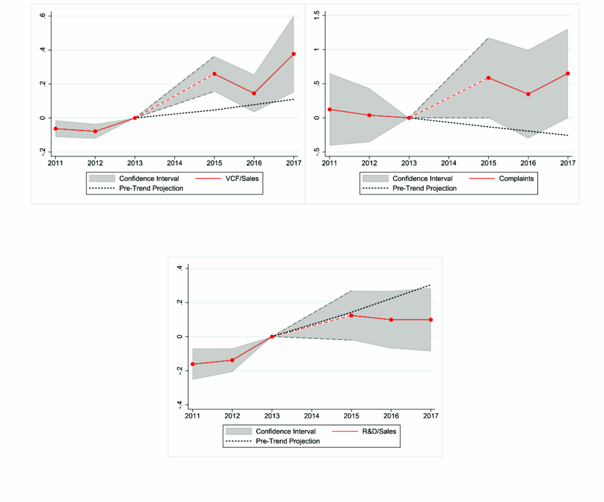

Figure 5: competition and R&D

Competition and patent protection (ModernBERT).

The table displays the results from entropy-balanced panel data regressions in which competition variables are the dependent variables. In columns (1)–(2), the dependent variable, *VCF/Sales*, is the measure of VC entry into a given firm's product market and is the total first-round dollars raised by the 25 startups from Venture Expert whose Venture Expert business description most closely matches the 10-K business description of the focal firm (using cosine similarities) in year *t*, scaled by focal firm sales in year *t-1*. Complaints is the number of paragraphs in the firm's 10-K that complain about competition divided by the total number of paragraphs in the firm's 10-K. R&D/Sales is R&D expenses in year *t* scaled by sales in year *t-1*. *Treatment* is the relative impact of the Alice decision on the firm's patent portfolio measured using KPSS dollar values or citations scaled by sales (see Eq. (2) for the formula). The *Treatment* was normalized by subtracting the mean and dividing by the standard deviation, resulting in a z-score transformation. In the odd and even-numbered columns, respectively, we use the KPSS and the number of citations approach to compute the *Patent Value* for the treatment variable. *Small* is a binary variable equal to one if a firm's total assets are less than the median total asset of its TNIC peers in 2013 and is zero otherwise. *Large* is 1-*Small*. *Post* is a dummy variable that equals one if the year is after the Alice decision and zero otherwise. *Small-Large Difference* is the difference between the estimates of Small × Post × Treatment and Large × Post × Treatment. All variables are described in detail in the variable list in Appendix A. All regressions include firm and year fixed effects. Standard errors are clustered at the firm level. *t*-statistics are reported in parentheses; *, **, and *** denote the significance at the level of 10%, 5%, and 1%.

Dependent variable:	$\frac{VCF}{Sales_{t-1}}$		Complaints		$\frac{R\&D}{Sales_{t-1}}$	
	(1)	(2)	(3)	(4)	(5)	(6)
Small × Post × Treatment	0.305***	0.113***	0.305*	0.175	0.200***	0.075**
	(5.43)	(2.94)	(1.94)	(1.66)	(2.94)	(2.16)
Large × Post × Treatment	0.002***	0.003***	−0.191	−3.775	−0.012**	−0.018
	(2.91)	(3.20)	(−1.58)	(−1.41)	(−2.22)	(−1.40)
Small-Large Difference	0.304***	0.110***	0.496**	3.949	0.211***	0.093**
	(5.40)	(2.85)	(2.50)	(1.47)	(3.10)	(2.51)
Observations	19 725	19 725	19 727	19 727	19 808	19 808
Firm Fixed Effects	YES	YES	YES	YES	YES	YES
Year Fixed Effects	YES	YES	YES	YES	YES	YES
Treatment Calculation	KPSS	Citation	KPSS	Citation	KPSS	Citation
Adj. R^2	0.153	0.061	0.008	0.043	0.091	0.025

Table 8: competition and R&D regressions

Table 9

Firm level competition and encroachment.

The table displays firm-pair-year panel data regressions in which pairwise product market encroachment (Delta TNIC Score) is the dependent variable. Delta TNIC Score is computed as the change in pairwise TNIC similarity (see Hoberg and Phillips 2016) from year *t-1* to year *t*. A large value indicates increased similarity and product market encroachment. To compute the RHS variables, we first sort firms into above and below median sales (relative to TNIC-2 peers) in 2013. We refer to the firms above and below the median as large and small, respectively. The variable Treat × Post is the sum of the Alice Scores for the two firms in the pair. Analogously, Large × Treat × Post is the sum of Treat × Post when either firm in the pair is large and is zero otherwise. Small × Treat × Post is analogously defined for when either firm in the pair is small. The last four variables focus on the properties of the pair. Large × Large × Treat Large × Post is the sum of Treat × Post for both firms in the pair when both are large (and is zero otherwise). Small × Small × Treat Small × Post is analogously defined when both are small. Large × Small × Treat Large × Post is Treat x Post for the large firm in the pair when the pair has one large firm and one small firm. Small × Large × Treat Small × Post is analogously defined using the small firm's treatment. We note that all level effects and Smaller-order interactions are subsumed by the fixed effects and, thus, are not reported. All regressions include firm-pair and year fixed effects and standard errors are clustered at the firm-pair level. *t*-statistics are reported in parentheses; *, **, and *** denote the significance at the level of 10%, 5%, and 1%.

Dependent variable:	Delta TNIC score		
	(1)	(2)	(3)
Treat × Post	0.0049***		
	(7.19)		
Large × Treat × Post		−0.0116***	
		(−13.17)	
Small × Treat × Post		0.0283***	
		(28.00)	
Large × Large × Treat Large × Post			−0.0124***
			(−10.17)
Large × Small × Treat Large × Post			−0.0105***
			(−8.59)
Small × Large × Treat Small × Post			0.0254***
			(19.34)
Small × Small × Treat Small × Post			0.0313***
			(20.51)
Observations	6,696,982	6,696,982	6,696,982
Pair Fixed Effects	YES	YES	YES
Year Fixed Effects	YES	YES	YES

6.7 Figure 6, Table 10, and Table 11: performance and barriers to entry

Read:

- Table 10 shows that large exposed firms gain in sales growth and profitability, while small exposed firms lose profitability and market value.
- Small firms' gross profit margins fall by about 43.7% relative to their pre-Alice mean, and market-to-book falls by about 12.5%.

- Figure 6 shows a particularly strong structural break in market-to-book valuations for small versus large firms.
- Table 11 shows that firms with low barriers to entry are hurt more, while firms with high non-redeployability, tangibility, and market share are better protected.
- Cash does not explain the same pattern as strongly as barriers to entry.

Economic message: The central mechanism is not simply financial resources. The important protection for large firms is that non-patent barriers to entry preserve rents after patent protection weakens.

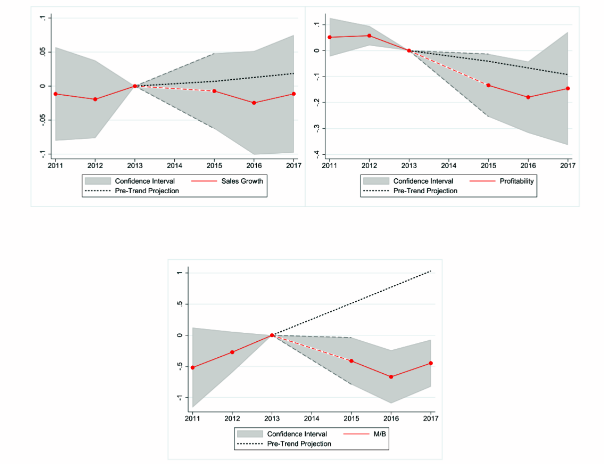

Figure 6: performance and valuation

Table 10
Profitability (ModernBERT).
The table displays the results from entropy-balanced panel data regressions that examine whether the profitability of large and small firms are differently affected by the Alice decision. In columns (1)–(2), the dependent variable is sales growth, calculated as the natural logarithm of total sales in the current year *t* divided by total sales in the previous year *t-1*; and in columns (3) and (4), Gross Profit Margin is sales minus cost of goods sold scaled by lagged sales. In columns (5)–(6), the dependent variable is the M/B Ratio, calculated as the market-to-book ratio (market value of equity plus book debt and preferred stock, all divided by book assets). *Treatment* is the relative impact of the Alice decision on the firm's patent portfolio measured using KPSS dollar values or citations scaled by sales (see Eq. (2) for the formula). The *Treatment* was normalized by subtracting the mean and dividing by the standard deviation, resulting in a z-score transformation. In the odd and even-numbered columns, respectively, we use the KPSS and the number of citations approach to compute the *Patent Value* for treatment. *Small* is a binary variable equal to one if a firm's total assets are less than the median total asset of its TNIC peers in 2013 and is zero otherwise. *Large* is 1-*Small*. *Post* is a dummy variable that equals one if the year is after the Alice decision and zero otherwise. *Small-Large Difference* is the difference between the estimates of Small × Post × Treatment and Large × Post × Treatment. All variables are described in detail in the variable list in Appendix A. All regressions include firm and year fixed effects. Standard errors are clustered at the firm level. *t*-statistics are reported in parentheses; *, **, and *** denote the significance at the level of 10%, 5%, and 1%.

Dependent variable:	Sales growth		Gross profit margin		M/B ratio	
	(1)	(2)	(3)	(4)	(5)	(6)
Small × Post × Treatment	0.013	0.002	−0.175***	−0.082**	−0.305***	−0.150*
	(0.72)	(0.18)	(−3.12)	(−2.48)	(−3.79)	(−1.68)
Large × Post × Treatment	0.015***	0.015**	0.019**	0.026***	−0.053	0.128
	(3.63)	(1.99)	(2.03)	(2.60)	(−0.80)	(0.88)
Small-Large Difference	−0.002	−0.013	−0.191***	−0.106***	−0.252**	−0.278*
	(−0.10)	(−1.02)	(−3.38)	(−3.10)	(−2.41)	(−1.62)
Observations	19 808	19 808	19 808	19 808	19 309	19 309
Firm Fixed Effects	YES	YES	YES	YES	YES	YES
Year Fixed Effects	YES	YES	YES	YES	YES	YES
Treatment Calculation	KPSS	Citation	KPSS	Citation	KPSS	Citation

Table 10: sales, profitability, and valuation

Table 11
Barriers to entry and financial liquidity (ModernBERT).
The table displays the results from entropy-balanced panel data regressions where we explore subsample results relating to barriers to entry and financial liquidity using models analogous to our main regressions for *Small* and *Large*. We report results for multiple dependent variables specified in the first column. We only report results for our treatment variable interacted with post and a high and low dummy for each subsample (specified in the headers over each column). For each regression, we thus report these two primary treatment effect coefficients. In the first two columns of results, the low and high are indicator variables based on below and above median assets, respectively, as in our baseline models (these are re-displayed here for comparison). The remaining columns report results in similar pairs and define low and high based on below or above median values of barriers to entry measures, including Asset Redeployability from Kim and Kung (2017) and PPEGT/Assets for low and high levels of tangible assets. Market Share is based on sales and the TNIC-2 industry classification. Regarding liquidity, we define cash as Compustat cash and short-term investments (che) scaled by assets. Due to space restrictions, we only show the results for the KPSS treatment variable. All regressions include controls and firm and year fixed effects, and standard errors are clustered by firm. *t*-statistics are reported in parentheses; *, **, and *** denote the significance at the level of 10%, 5%, and 1%.

Dependent variable:	Assets (Baseline)			Non-Redeployability High = BTEs Stronger			PPEGT/Assets High = BTEs Stronger			Market Share High = BTEs Stronger			Cash/Assets High = More Liquidity		
	Small	Large	S-L	Low	High	L-H	Low	High	L-H	Low	High	L-H	Low	High	L-H
Patent App./Assets	−0.006***	−0.002***	***	−0.005***	−0.007**		−0.006***	−0.004***		−0.007***	−0.001***	***	−0.003	−0.006***	
	(−7.32)	(−2.71)		(−5.60)	(−2.51)		(−6.10)	(−2.63)		(−6.79)	(−2.62)		(−1.50)	(−6.00)	
Patent App./Sales	−0.015***	−0.002**	***	−0.012***	−0.013**		−0.014***	−0.007***	**	−0.015***	−0.002***	***	−0.006	−0.013***	
	(−9.79)	(−2.07)		(−8.12)	(−2.33)		(−8.69)	(−2.72)		(−10.06)	(−2.80)		(−1.43)	(−8.85)	
VC Investments	0.305***	0.002***	***	0.323***	0.039	***	0.316***	0.044	***	0.330***	−0.002*	***	0.085	0.296***	**
	(5.43)	(2.91)		(4.97)	(0.90)		(4.86)	(1.37)		(5.21)	(−1.83)		(1.62)	(4.53)	
Complaints	0.295*	−0.186	**	0.154	0.076		0.101	0.055		0.147	−0.085		0.101	0.088	
	(1.77)	(−1.46)		(0.89)	(0.39)		(0.62)	(0.23)		(0.83)	(−0.52)		(0.57)	(0.53)	
Nondisclosure	0.006***	0.000	***	0.004**	0.000	**	0.003*	0.003		0.004**	0.000	*	0.001	0.003*	
	(2.68)	(0.04)		(2.21)	(0.07)		(1.93)	(1.51)		(2.08)	(0.85)		(1.27)	(1.92)	
RD/Sales	0.200***	−0.012**	***	0.202**	−0.003	**	0.197**	0.020	**	0.207**	−0.003	**	0.067*	0.177**	
	(2.94)	(−2.22)		(2.46)	(−0.18)		(2.35)	(0.60)		(2.43)	(−1.01)		(1.69)	(2.20)	
Sales Growth	0.013	0.015***		0.013	0.019***		0.005	0.030***		0.014	0.015***		0.027***	0.013	
	(0.72)	(3.63)		(0.60)	(2.62)		(0.22)	(2.81)		(0.59)	(2.80)		(2.93)	(0.60)	
Profitability	−0.172***	0.019**	***	−0.185***	0.046***	***	−0.185***	0.005	**	−0.183***	0.019*	***	−0.011	−0.162**	**
	(−3.08)	(2.02)		(−2.76)	(4.03)		(−2.71)	(0.14)		(−2.65)	(1.92)		(−0.29)	(−2.48)	
Tobins'Q	−0.305***	−0.053	**	−0.295***	−0.155		−0.298***	−0.204**		−0.322***	−0.047	**	−0.017	−0.313***	***
	(−3.79)	(−0.80)		(−3.38)	(−1.57)		(−3.43)	(−2.19)		(−3.80)	(−0.48)		(−0.30)	(−3.71)	

6.8 Table 12 and Table A.13: lawsuits, nondisclosure, and variable definitions

Read:

- Table 12 shows that small exposed firms face more NPE lawsuits and alleged infringement suits.
- Large exposed firms face fewer NPE and operating-company lawsuits after Alice.
- Small firms increase nondisclosure-agreement language in 10-Ks, consistent with substitution toward secrecy after patent protection weakens.
- Table A.13 defines core variables, including the Alice ModernBERT score, treatment, R&D/sales, patent applications, lawsuits, nondisclosure, VC entry, and competition complaints.

Economic message: Small firms appear to lose legal protection and try to replace it through secrecy contracts, while large firms benefit from a reduction in patent-lawsuit threats.

Lawsuits (ModernBERT).
The table displays the results from entropy-balanced panel data regressions examining whether lawsuit frequencies of large and small firms were differently affected by the Alice decision. *Num. of NPE Alleged* is the total number of lawsuits that the firm was alleged by a non-practicing entity in that year. *Num. of OC Alleged* is the number of lawsuits that the firm was alleged by an operating company in that year. *Number of Alleged* is the total number of lawsuits that the firm was alleged in that year. *Nondisclosure* is the total number of 10-K words mentioning "non-disclosure", "NDA agreements", "confidentiality agreement" or "secrecy agreement" scaled by the total number of words in the 10-K. *Treatment* is the relative impact of the Alice decision on the firm's patent portfolio measured using KPSS dollar values or citations scaled by sales (see Eq. (2) for the formula). The *Treatment* was normalized by subtracting the mean and dividing by the standard deviation, resulting in a z-score transformation. In the odd and even-numbered columns, respectively, we use the KPSS and the number of citations approach to compute the *Patent Value* for treatment. *Small* is a binary variable equal to one if a firm's total assets are smaller than the median total asset of its TNIC peers in 2013, and it is zero otherwise. *Large* is 1-*Small*. *Post* is a dummy variable that equals one if the year is after the Alice decision and zero otherwise. *Small-Large Difference* is the difference between the estimates of Small × Post × Treatment and Large × Post × Treatment. All variables are described in detail in the variable list in Appendix A. All regressions include firm and year fixed effects. Standard errors are clustered at the firm level. *t*-statistics are reported in parentheses; *, **, and *** denote the significance at the level of 10%, 5%, and 1%.

Dependent variable:	Num. of NPE Alleged		Num. OC Alleged		Number of Alleged		Nondisclosure	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Small × Post × Treatment	0.008**	−0.002	0.007	−0.002	0.017**	−0.003	0.006***	0.002**
	(2.14)	(−0.33)	(1.27)	(−0.28)	(2.18)	(−0.37)	(2.67)	(2.50)
Large × Post × Treatment	−0.063**	−0.182**	−0.053***	−0.171***	−0.124***	−0.455***	−0.000	−0.001*
	(−2.25)	(−2.13)	(−2.63)	(−3.41)	(−2.91)	(−5.07)	(−0.04)	(−1.77)
Small-Large Difference	0.072**	0.180**	0.060***	0.170***	0.142***	0.452***	0.006***	0.003***
	(2.52)	(2.10)	(2.87)	(3.36)	(3.27)	(5.01)	(2.58)	(3.06)
Observations	19 808	19 808	19 808	19 808	19 808	19 808	18 486	18 486
Firm Fixed Effects	YES	YES	YES	YES	YES	YES	YES	YES
Year Fixed Effects	YES	YES	YES	YES	YES	YES	YES	YES
Treatment Calculation	KPSS	Citation	KPSS	Citation	KPSS	Citation	KPSS	Citation
Adj. R^2	0.012	0.021	0.018	0.028	0.015	0.034	0.045	0.021

Table 12: lawsuits and NDAs

Variable Definitions.

Variable	Definition
Panel A: Firm Characteristics	
Assets	Compustat item AT
Sales	Compustat item SALE
Cash	Compustat item CHE
Gross Profit Margin	Compustat items Sales minus COGS divided by lagged total sales.
M/B Ratio	Compustat sum of market equity (CSHO * $PRCC_F$), DLC, DLTT, PSTKL, all scaled by lagged book assets.
Sales Growth	Natural logarithm of total sales in the current year t divided by total sales in the previous year t-1.
Log(Age)	Natural logarithm of one plus the current year of observation minus the first year the firm appears in the Compustat database.
Asset Redeployability	Calculated using the procedure from Kim and Kung (2017).
PPEGT	Compustat item Property Plant and Equipment, Gross Total
Market Share	Firm sales/total industry sales using the TNIC-2 industry classification.
Panel B: Innovation, Acquisition & Lawsuit Characteristics	
Alice ModernBERT Score	Alice ModernBERT Score is the predicted probability that the patent would be rejected based on the Alice decision using the ModernBERT LLM.
Treatment	Treatment is a weighted average of a firm's patent values multiplied by the Alice ModernBERT Score scaled by sales. For a firm's patent portfolio, we gather all patents that are valid by the third quarter of 2014. Firm's patent value is calculated in two ways: i) dollar amount provided by KPSS; ii) citations that the patent received. The mathematical notation is provided in Eq. (2).
$R\&D/Sales_{t-1}$	Compustat XRD divided by lagged total sales, set to zero if XRD is missing.
Num. Patent App.	The number of patent applications.
Num. Alleged	It is the number of lawsuits that the firm was alleged for infringing on a patent.
Num. NPE Alleged	It is the number of lawsuits that the firm was alleged by an NPE for infringing on a patent.
Num. OC Alleged	It is the number of lawsuits that the firm was alleged by an OC for infringing on a patent.
Nondisclose	Total number of 10-K words mentioning "non-disclosure", "NDA agreements", "confidentiality agreement" or "secrecy agreement" scaled by the total number of words in the 10-K.
Panel C: Competition Measures	
$VCF/Sales_{t-1}$	A measure of VC entry in a given firm's product market computed as the total first-round dollars raised by the 25 startups from Venture Expert whose Venture Expert business description is most similar to the 10-K business description of the focal firm (using cosine similarities), scaled by focal firm lagged sales.
Complaints	The number of paragraphs in the firm's 10-K that mention competition divided by the total number of paragraphs in the firm's 10-K.

Table A.13: variable definitions

7 Short summary

- The paper studies the consequences of weakened IP protection after the 2014 Alice Supreme Court decision.
- It uses ModernBERT to score hundreds of thousands of patents for Alice exposure.
- ModernBERT outperforms traditional NLP models and predicts Alice references in post-grant reviews.
- About one-fifth of patents in the impacted sample are predicted to lose protection.
- Exposed firms reduce patenting after Alice.
- Small exposed firms suffer more competition, product-market encroachment, lower profits, and lower valuation.
- Large exposed firms often benefit through higher sales growth, higher profitability, and reduced lawsuits.
- Barriers to entry explain why large firms are better insulated.
- Small firms respond by increasing R&D and nondisclosure language.
- The paper shows that weakening patents can shift rents from smaller innovators toward large incumbents.

8 Conclusion

8.1 Core conclusion

Acikalin, Caskurlu, Hoberg, and Phillips show that weakening patent protection has sharply heterogeneous effects. Alice reduces patenting and weakens some firms' IP protection, but the economic impact depends on firm size and non-patent barriers to entry. Large exposed firms can gain, while small exposed firms lose profits and valuation.

8.2 Economic intuition

Patents are not only monopoly rights. They can also protect smaller firms from large incumbents. When patent protection weakens, small firms lose a legal barrier that helped preserve their niches. Large firms can rely on scale, product breadth, asset specificity, and market share. As a result, the loss of formal IP protection may increase entry into small-firm product spaces while leaving large-firm rents protected or even enhanced.

8.3 Takeaway

The paper's main lesson is that patent policy has distributional consequences across firms. Weakening patents may reduce some legal barriers, but it can also undermine small-firm protection and shift market power toward large firms. The effect of IP policy therefore depends not only on patents themselves but also on firms' non-patent competitive advantages.